

JAN SAMADHAN: REAL-TIME PUBLIC COMPLAINT PRIORITIZATION USING AI

Capstone-I project report submitted

by the student of

RONIT KUMAR

Hybrid UG program in Computer Science & Data Analytics

Student Name :- Ronit Kumar

Roll No :- 2312res_541

Group No :- 59

INDIAN INSTITUTE OF TECHNOLOGY PATNA

BIHTA - 801106, INDIA

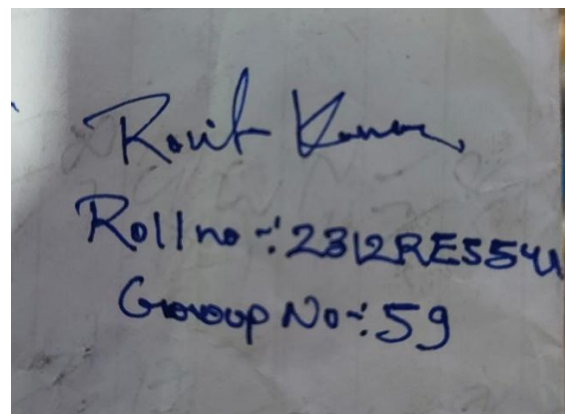
Date:-25/03/2026

Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Date:-20/03/2026

Student Name , Roll No , Group No and Signature



Handwritten signature: Ravit Kumar
Roll no :- 2312RES54
Group No :- 59

SUMMARY OF THE PROJECT

In many cities and towns, people face difficulties when they try to report everyday problems such as damaged roads, water leakage, electricity failure, or garbage collection issues.

Although some government portals exist, complaints are often handled slowly because there is no proper system to identify which issues are more urgent. As a result, serious problems sometimes remain unresolved for a long time.

The Aim of this project is to design and develop a smart complaint management system that can automatically understand and prioritize public complaints. The system allows users to submit their complaints through an online platform. Once a complaint is submitted, it is processed using a simple intelligent model that analyzes the text and identifies the type of issue. Based on the content, the system assigns a priority level such as high, medium, or low.

For example, complaints related to accidents, medical emergencies, or public safety are marked as high priority, while issues like street light repair or garbage collection are treated as medium or low priority depending on the situation. This helps ensure that urgent problems receive faster attention from the concerned authorities.

The system also organizes complaints based on categories like transportation, sanitation, water supply, and electricity. All complaints are stored in a database and displayed on an admin

dashboard, where officials can easily view and manage them. High-priority complaints are highlighted so that they can be resolved quickly.

The project is implemented using a combination of web development and basic machine learning techniques. The frontend provides a user-friendly interface for submitting complaints, while the backend handles data processing and model predictions. The overall goal is to make the complaint handling process more efficient, transparent, and responsive. This project can be useful for local authorities, municipal bodies, and government organizations to improve public service delivery. It also demonstrates how technology can be used to solve real-life problems and make communication between citizens and authorities more effective.

TEAM WORK

Ronit Kumar (Team Lead & AI Specialist): Collected and cleaned data, performed EDA, preprocessed text, built the ML model, and guided the project development.

Rohit kumar (Frontend Developer): Designed and implemented the user interface, complaint forms, and dashboards.

Rohit kumar (Backend Developer): Developed APIs, integrated the AI model, and managed server-side logic.

Sachin kumar (Database & Deployment Specialist): Handled database design, storage, and deployment of the system.

Sachin kumar (Testing & Documentation): Tested the system for functionality and performance and prepared project documentation.

CONTENTS

A. Introduction

B. Chapter:

1. Data Collection
2. Exploratory Data Analysis (EDA)
3. NLP Preprocessing
4. Machine Learning Model Development

C. Conclusion & future work

D. References

Introduction

In today's fast-growing urban and semi-urban environments, the number of public issues faced by citizens has increased significantly. Problems such as damaged roads, water leakage, electricity failures, improper waste management, and public safety concerns are common in many areas. Although government authorities provide platforms for citizens to report these issues, the process of handling complaints is often slow and inefficient. One of the main reasons for this delay is the lack of a proper system to identify and prioritize complaints based on their urgency and impact.

In most traditional complaint management systems, all complaints are treated equally and processed in the order in which they are received. This approach creates a serious problem because urgent complaints, such as road accidents, medical emergencies, or public safety threats, may not receive immediate attention. At the same time, less critical issues may be handled earlier simply because they were submitted first. This results in delays in resolving critical problems and reduces the overall effectiveness of public service systems. Moreover, manual handling of complaints requires significant human effort, increases the chances of errors, and makes the system less transparent.

To address these challenges, there is a need for an intelligent system that can automatically understand the nature of complaints and prioritize them accordingly. The **JAN SAMADHAN: Real-Time Public Complaint Prioritization Using AI** project is designed with this objective in mind. The main goal of this system is to improve the efficiency and responsiveness of complaint handling by using machine learning and natural language processing techniques. The system provides an online platform where citizens can easily submit their complaints in the form of text. Once a complaint is submitted, it is automatically analyzed by the system. Using natural language processing, the system understands the content of the complaint and identifies the type of issue being reported. For example, a complaint mentioning a road accident will be categorized under transportation, while a complaint about garbage collection will fall under sanitation. This automatic classification helps in organizing complaints and directing them to the appropriate department.

In addition to classification, the system also assigns a priority level to each complaint. This is one of the most important features of the project. The priority is determined based on the urgency and seriousness of the issue described in the complaint. For instance, complaints related to accidents, health emergencies, or safety risks are marked as high priority, while routine issues such as street light repairs or minor

maintenance are categorized as medium or low priority. By doing this, the system ensures that critical problems are addressed quickly, while less urgent issues are handled in a structured manner.

Another important aspect of the project is the use of machine learning models to automate the decision-making process. Instead of relying on manual judgment, the system learns patterns from previously collected complaint data. This allows it to make consistent and accurate predictions for new complaints. The use of such technology reduces human effort, improves speed, and minimizes errors in classification and prioritization.

The project also includes a backend system that stores all complaints and their corresponding details in a structured database. An admin dashboard is provided for officials to view and manage complaints efficiently. Through this dashboard, authorities can monitor high-priority issues, track complaint status, and take necessary actions. This not only improves the workflow but also increases transparency in the system.

The significance of this project lies in its practical application. It addresses a real-world problem that affects a large number of people and provides a scalable solution that can be implemented by municipal bodies and government organizations. By integrating artificial intelligence with public service systems, the project demonstrates how technology can be used to improve governance and enhance the quality of life for citizens.

CHAPTER 1

Data Collection

The first step in my role was to assemble a comprehensive and high-quality dataset. Since the JAN SAMADHAN system is designed to prioritize public complaints, the data needed to represent a variety of real-world civic issues, such as transportation, sanitation, electricity, water supply, healthcare, and public safety. I collected data from multiple sources:

1. **Open Government Data Portals:** Publicly available datasets on civic grievances and complaints were accessed from government websites such as data.gov.in. These datasets contained historical complaints submitted by citizens across various categories.
2. **Municipal and Local Portals:** Certain city-specific portals provided records of local complaints. These were used to understand real citizen reporting patterns.
3. **Synthetic Data Generation:** To ensure sufficient data for training machine learning models, I created additional complaint examples. This allowed representation of multiple categories and priority levels, ensuring the model could generalize well to unseen complaints.

The dataset was structured with the following fields:

- **complaint_text**: The textual description of the problem
- **category**: The department or type of complaint (e.g., Transport, Sanitation, Water Supply)
- **priority**: Urgency level (High, Medium, Low)
- **location and timestamp** (for future enhancement)

This dataset was saved in CSV format, with approximately 3000–5000 rows to provide sufficient training examples for the machine learning model.

```
df = pd.read_csv("/content/drive/MyDrive/smart_complaints_dataset_4000.csv")  
df.head()
```

...

	complaint_id	complaint_text	category	priority	department	location	date_submitted	status
0	1	Noise disturbance at night	Police/Safety	Medium	Police/Safety	Kolkata	2025-03-25	In Progress
1	2	Overflowing dustbins in the area	Sanitation	High	Sanitation	Chennai	2025-07-04	Pending
2	3	Shortage of medicines in health center	Health	Medium	Health	Ranchi	2024-08-14	Resolved
3	4	Vaccination not available in local clinic	Health	Medium	Health	Mumbai	2024-06-09	In Progress
4	5	UPI payment fraud reported	IT Fraud	Medium	IT Fraud	Hyderabad	2025-02-19	Resolved

CHAPTER 2

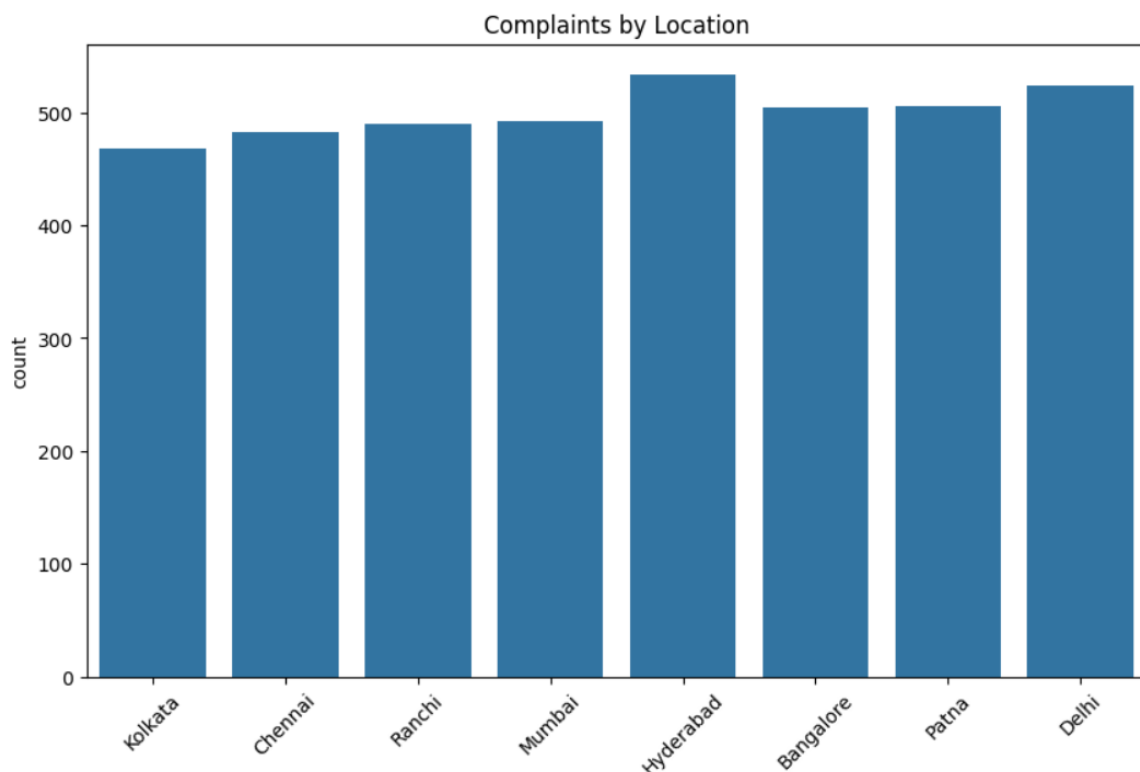
Exploratory Data Analysis (EDA)

After collecting the data, I performed **Exploratory Data Analysis (EDA)** to understand its structure, distribution, and characteristics. EDA is critical because it reveals patterns, detects anomalies, and guides the feature engineering process.

Key steps included:

1. Data Overview:

- Counted the total number of complaints
- Checked for missing values
- Identified duplicate entries



2. Category Distribution:

- Visualized the number of complaints per category using bar charts
- Observed which categories occurred most frequently to identify potential bias

3. Priority Distribution:

- Checked how complaints were distributed across priority levels
- Ensured there was a balanced representation of high, medium, and low priority complaints

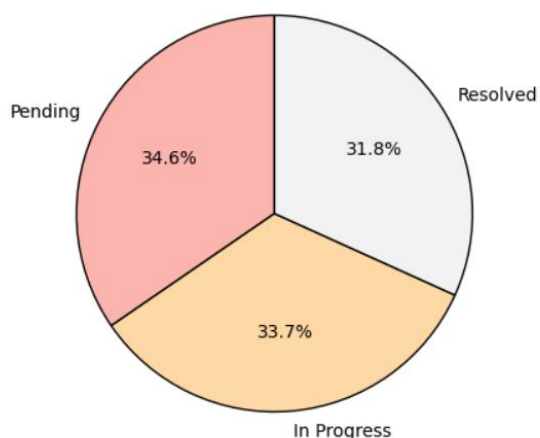
4. Text Length Analysis:

- Analyzed the number of words per complaint
- Helped understand if preprocessing methods like tokenization or truncation were necessary

EDA helped me prepare the dataset for **text preprocessing** and informed the choice of features for the machine learning model.

```
plt.figure(figsize=(5, 5))
df['status'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90, cmap='Pastel1', wedgeprops={'edgecolor': 'black'})
plt.title('Distribution of Complaint Statuses')
plt.ylabel('')
plt.show()
```

Distribution of Complaint Statuses



CHAPTER 3

NLP Preprocessing

Since complaints are written in **natural language**, preprocessing was essential to convert raw text into a format suitable for machine learning. The steps involved were:

1. **Lowercasing:**

All text was converted to lowercase to ensure uniformity. For example, “Road Accident” and “road accident” are treated the same.

2. **Removing Punctuation and Special Characters:**

Symbols such as !@# were removed because they do not contribute to the meaning of the complaint.

3. **Tokenization:**

Complaints were split into individual words or tokens using **NLTK**. This step allows algorithms to process the text as a sequence of words.

4. **Stopword Removal:**

Common words like “the,” “is,” and “and” were removed because they do not help in identifying the complaint type or priority.

5. **Lemmatization:**

Words were reduced to their base form (e.g., “leaking” → “leak”) to group similar words together and reduce feature dimensionality.

6. **Vectorization:**

The preprocessed text was converted into numerical representations suitable for machine learning. Two common approaches were considered:

- **TF-IDF (Term Frequency–Inverse Document Frequency):** Captures the importance of words across complaints. Rare but meaningful words are given higher weights.
- **Bag-of-Words (BoW):** Represents text as a frequency vector of words.

This preprocessing pipeline ensured that the AI model could accurately learn patterns from the complaint text.

```
▶ y_priority = df["priority"]

X_train,X_test,y_train,y_test = train_test_split(
    X,y_priority,test_size=0.2,random_state=42
)

model_priority = LogisticRegression(max_iter=200)

model_priority.fit(X_train,y_train)

pred_pri = model_priority.predict(X_test)

print("Accuracy:",accuracy_score(y_test,pred_pri))

print(classification_report(y_test,pred_pri))
```

```
... Accuracy: 0.37625
```

	precision	recall	f1-score	support
High	0.28	0.18	0.22	232
Low	0.32	0.15	0.20	243
Medium	0.41	0.69	0.52	325
accuracy			0.38	800
macro avg	0.34	0.34	0.31	800
weighted avg	0.35	0.38	0.34	800

CHAPTER 4

Machine Learning Model Development

The main goal of the AI model is to **classify complaints into categories** and **predict their priority**. I adopted a **two-step approach**:

Step 1: Complaint Category Classification

- **Problem Type:** Multi-class classification
- **Input:** Preprocessed complaint text
- **Output:** Complaint category (e.g., Transport, Sanitation, Water Supply)

Algorithms Used:

- **Logistic Regression:** Suitable for text classification, provides interpretable probabilities.
- **Random Forest:** An ensemble method combining multiple decision trees to improve accuracy.
- **Support Vector Machine (SVM):** Effective for high-dimensional data such as TF-IDF vectors.

Training Process:

- Split data into training (80%) and testing (20%) sets
- Vectorized complaint text using TF-IDF
- Trained multiple models and evaluated accuracy using metrics such as **precision, recall, F1-score, and confusion matrix**

The model with the best balance of accuracy and interpretability was selected for deployment.

Step 2: Priority Prediction

- **Problem Type:** Multi-class classification (High, Medium, Low)
- **Approach:**

- Initially, simple **keyword-based heuristics** were used to tag high-priority complaints. Keywords like “accident,” “fire,” “hospital,” and “emergency” indicated urgency.
- Then, a machine learning model (Random Forest or Logistic Regression) was trained on the labeled dataset to learn patterns beyond simple keywords.
- **Evaluation:**
 - Confusion matrices and classification reports were used to check whether high-priority complaints were correctly predicted.
 - Priority predictions were cross-checked with category predictions to ensure consistency.

```
def predict_complaint(text):  
  
    clean = clean_text(text)  
  
    vec = vectorizer.transform([clean])  
  
    category = model_category.predict(vec)[0]  
  
    priority = model_priority.predict(vec)[0]  
  
    print("Category:",category)  
    print("Priority:",priority)  
  
predict_complaint("UPI payment fraud reported")
```

```
Category: IT Fraud  
Priority: Medium
```

CONCLUSION & FUTURE WORK

It was a wonderful learning experience for me while working on this project. This project took me through the various phases of project development and gave me real insight into the world of Computer science ,data science and the web development field. The joy of work with my teammates and the thrill involved and challenges gave me a feel of developer industry.

I enjoyed each and every bit of work and I had a great experience with my teachers and teammates.

In future I will work on backend web development with this topic and we learn many other programming language like Figma , Node.js , React.js , Javascript. After that i learn Networking, Server, 'how to host a website',creating a API then we implement all this in Project.

Thank you

